

Chapter 2

Basic Concepts in Organic Chemistry

Delocalized Chemical Bonding

In the previous chapter we have seen that the electrons involved in the formation of a bond are shared by only two nuclei. Such bonds are referred to as *localized chemical bonds*. If the bonding electrons are shared by more than two nuclei in a molecule, the bonding is said to be *delocalized*. The molecules with delocalized chemical bonds contain one or more bonding molecular orbitals that are spread out over more than two nuclei. There are three main types of structures that show delocalization.

1. Double (or triple) bond in conjugation with another double (or triple) bond. Double (or triple) bonds separated by a single bond are said to be in conjugation with each other. The simplest molecule containing a conjugated system of double bonds is 1,3-butadiene, $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$. In this molecule, each carbon atom being bonded to three other atoms, uses sp^2 hybrid orbitals to form σ bonds, leaving a p orbital (containing one electron) on each carbon atom as such. In this way all the atoms (four carbons and six hydrogens) lie in one plane, and the p orbitals are perpendicular to the plane of the molecule and parallel to one another. Each p orbital is in a position to overlap equally with the adjacent p orbitals on either side. This mutual overlap of four p orbitals produces four new molecular orbitals (all having different energy) two of which are bonding that contain two electrons each, and two are antibonding which are vacant (Fig. 2.1).

The plane of the molecule is a nodal plane for each molecular orbital which thus has two parts, one above and the other below the plane. It can be seen that the bonding orbital of higher energy, i.e., π_2 , also has another node which is perpendicular to the nodal plane. Similarly, the antibonding orbitals π_3^* and π_4^* have two and three such nodes, respectively. In general, when p orbitals overlap, as in 1,3-butadiene, they do so in such a way as to produce one molecular orbital having no node, another molecular orbital with one node, third with two, and fourth with three nodes. These nodes are in addition to the nodal plane. The relative energy of the molecular orbitals depends on the number of nodes; the greater the number of nodes, the higher the energy of the orbital.

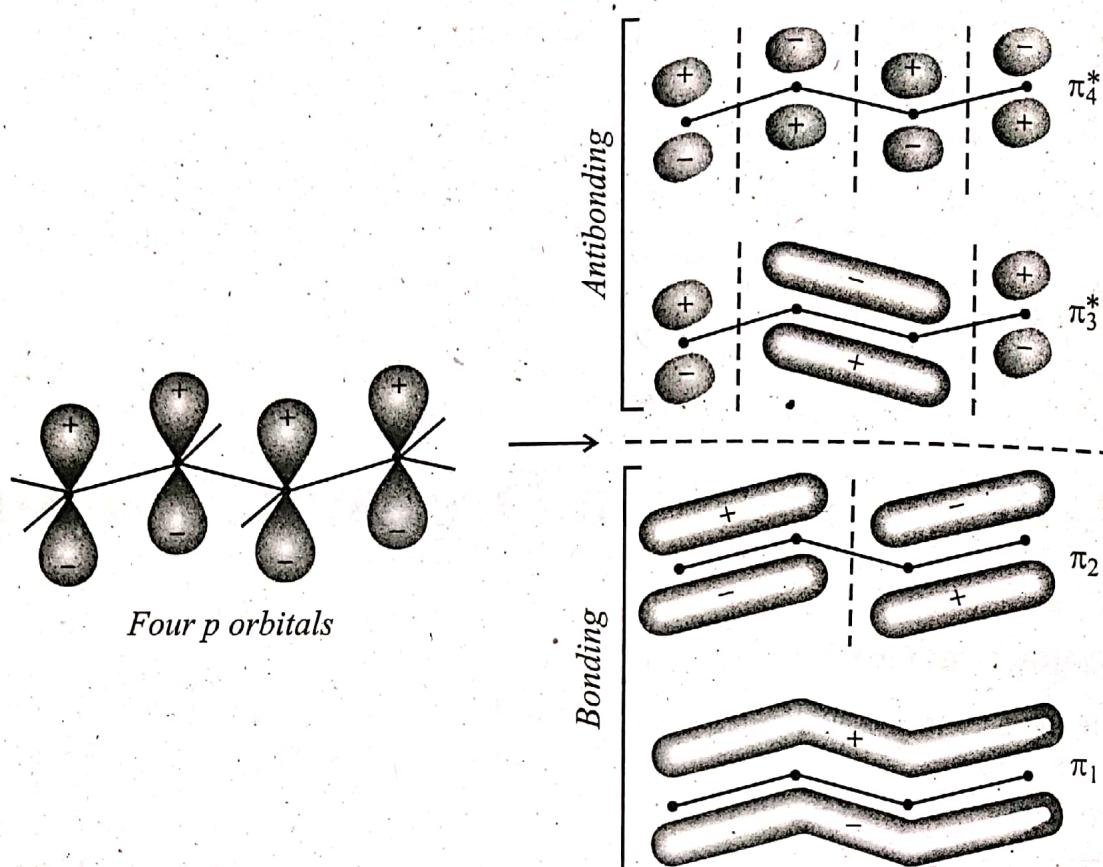


Fig. 2.1. Four π molecular orbitals of 1,3-butadiene, formed by the overlap of four parallel p orbitals.

Similar delocalization is found in other conjugated systems such as $\text{C}=\text{C}-\text{C}=\text{O}$ and $\text{C}\equiv\text{C}-\text{C}\equiv\text{N}$, in systems with three or more conjugated multiple bonds, and in systems where double or triple bonds are in conjugation with aromatic rings.

2. Double (or triple) bond in conjugation with a p orbital. In this case, a p orbital is in an atom which is linked to a double-bonded atom through a single bond as exemplified by an allylic system, $\text{CH}_2=\text{CH}-\overset{+, \text{ or } -}{\text{CH}_2}$. Since a double bond is formed by the overlap of two p orbitals, there are now three p orbitals in the allylic system, which are parallel to one another and perpendicular to the plane of the molecule. Since the overlap of n atomic orbitals produces n molecular orbitals, overlap of a p orbital with an adjacent double bond (two p orbitals) gives rise to three new molecular orbitals — a bonding, a nonbonding and an antibonding molecular orbital (Fig. 2.2).

As in the case of 1,3-butadiene, the plane of the allylic molecule is the nodal plane for all the molecular orbitals, and this is the only node in the bonding orbital. In the nonbonding orbital the middle carbon is not involved in the overlap so that another node is passing through this carbon, and the two remaining p orbitals are so far apart that overlap is very small making this molecular orbital almost nonbonding. The energy of the nonbonding orbital is equal to the average energy of the component p orbitals. The antibonding orbital has two such nodes in addition to the nodal plane.

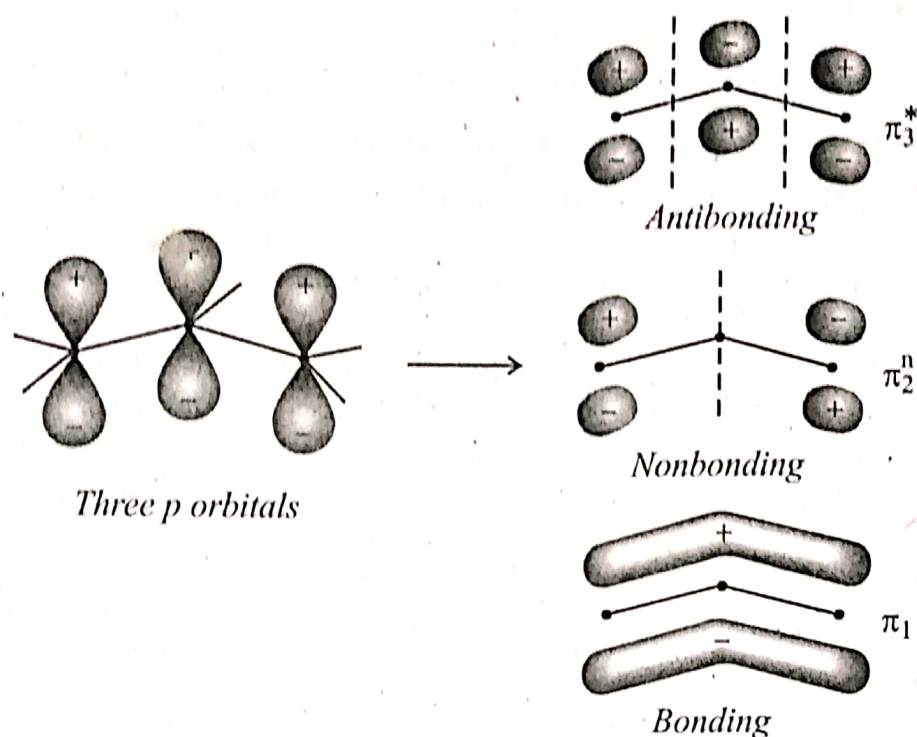


Fig. 2.2. Three π molecular orbitals of an allylic system, formed by the overlap of three parallel p orbitals.

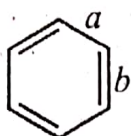
The original double bond of the allylic system contributes two electrons, whereas the original p orbital may contribute none, one or two electrons, depending on whether the allylic system is a cation, a radical or an anion. Thus, the total number of electrons to be accommodated by the new molecular orbitals is two, three or four. The two electrons of opposite spin in the allylic system (cation, radical or anion) occupy the bonding orbital, whereas the third electron of allyl radical must be put into the higher energy nonbonding orbital. The fourth electron of allyl anion can also be put into this second orbital. The antibonding orbital of the highest energy remains vacant in all these cases. Thus, the orbital structures of the allylic carbocation, free radical and carbanion differ from each other only in that the nonbonding orbital is empty, half-filled or completely filled, as shown below:

Molecular orbital	Cation	Radical	Anion
Antibonding, π_3^*	—	—	—
Nonbonding, π_2^n	—	1	↑↓
Bonding, π_1	↑↓	↑↓	↑↓

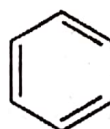
3. Sigma bond in conjugation with a double (or triple) bond or a p orbital. This type of system involves the delocalization of σ electrons and is discussed under *hyperconjugation* (see p. 40).

The Resonance Theory

Generally, the Lewis structure of a compound provides information about its behaviour. For example, a compound with the structure $\text{CH}_3\text{—CH}^b\text{=CH}^a\text{—CH}_2\text{OH}$ will add a molecule of bromine (due to carbon-carbon double bond) and will react with acetic anhydride to yield an ester (due to alcoholic group). The bond length of the bond *a* will be 1.54 Å and that of the bond *b* will be 1.33 Å. Similarly, if the properties of a compound are known, a Lewis structure can be drawn for it. But this is not true for a compound having a delocalised system of bonding, and the behaviour of such a compound sometimes differs so widely from what we expect from the Lewis structure ordinarily written for it that this structure becomes of little value and, in fact, does not seem to be the real structure of that compound. For example, the structure I ordinarily written for benzene is expected to add three molecules of bromine, but in actual practice it does not do so. Neither the bond length of the bond *a* is 1.54 Å and that of the bond *b* is 1.33 Å, rather both bond lengths are equal and are in between 1.54 Å and 1.33 Å.



I



II

In such cases we have to include some more structures so that all the structures together are in a better position to explain the behaviour of the compound, but none of them is the real structure of the compound. For example, the structure II and some others (III–V) are also considered along with the structure I to better explain the characteristics of benzene. All these structures may not be of equal importance and may differ in the extent of their contribution toward explaining the properties of the compound. The real structure of the compound is considered to be a *weighted average* of all the contributing structures. *The representation of a real structure as a weighted average of two or more contributing structures* is called **resonance**. The contributing structures are called **canonical forms** and the real structure is called **resonance hybrid**.

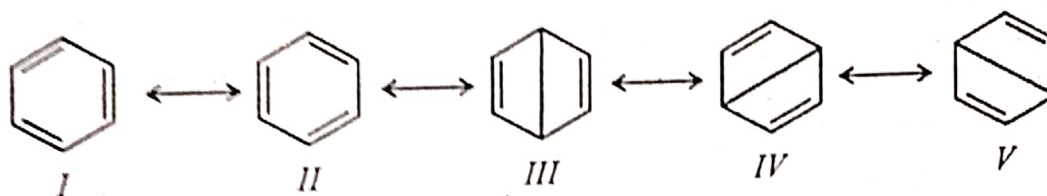
Sometimes, the actual structure of a molecule can only be expressed by making reference to certain Lewis structures (canonical forms) and assuming that the actual structure is a resonance hybrid of them all. The canonical forms have no real physical existence. On the other hand, the structure of the molecule which does exist physically is not really known to us. For example, there is no substance anywhere in the world having structure I, and the actual structure of benzene is not known.

Previously, we have used only the *molecular-orbital method of approximation* to solve the Schrodinger wave equation (Chapter I). There is another general method known as the *valence-bond method of approximation* for solving the wave equation, which is particularly useful when more than one structure are written to express the actual structure of a compound. In this method, various Lewis structures (canonical forms) that a molecule may possibly have, are drawn and a wave equation (ψ) is written for each of

them. The wave equation for the real structure (Ψ) is obtained by taking the sum of the wave equations of all the canonical forms with their corresponding weighting factors (C).

$$\Psi = C_A \psi_A + C_B \psi_B + C_C \psi_C + \dots$$

In this equation, each C represents the extent to which the corresponding canonical form contributes to the actual structure. Values for C are obtained by solving the equation for various values of each C and choosing the solution that gives lowest energy, as in the case of molecular-orbital method. In fact, both methods give similar solutions for molecules that contain only localized bonds. For compounds containing delocalized bonds, the valence-bond method becomes much more difficult to apply as the structures become more and more complicated, and in that case the molecular-orbital method is used much more often for the quantitative solution of wave equation. However, for the qualitative description of the structure of molecules, the valence-bond method is often used. The structure of benzene is best represented when canonical forms I–V with double headed arrows in between them, are put together. This is the resonance method of writing the structure of a molecule.

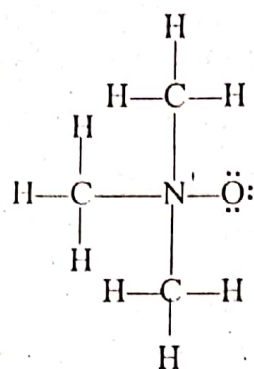


When the wave equation is solved by considering that all these canonical structures contribute to the resonance hybrid, the energy value obtained is lower than that of any of the canonical forms. The difference in energy between the resonance hybrid and the canonical form of lowest energy is called the **resonance energy**. The contribution of structures I and II to the actual structure has been calculated to be 39% each, whereas that of III, IV and V is 7.3% each. The carbon-carbon bond length is more than that of a double bond and less than that of a single bond, i.e., its character is in between a double bond and a single bond, but all the bonds have the same length.

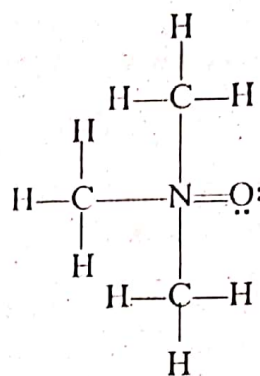
The resonance method of describing the structure of a molecule, as shown above in the case of benzene, does not mean that the structure rapidly shifts between the canonical forms, neither it means that some molecules have one canonical form and some have another. In fact, all the molecules of a substance have the same structure and for all the times, in which the electrons are delocalized over the whole system involved in the resonance.

Canonical forms. For drawing the canonical forms to express the actual structure of a molecule we are guided by certain rules which are as follows:

1. All the canonical forms must be bonafide Lewis structures. Each canonical form should follow the rules of valence, i.e., it should not have more than two electrons in the outer shell (valence shell) of first-row elements (e.g., H), more than eight electrons in the outer shell of second-row elements (e.g., C, N, O) and more than twelve electrons in the outer shell of third-row elements (e.g., P, S). Thus, the structure VI, not being a valid Lewis structure, cannot be considered as a canonical form of trimethylamine oxide.



Trimethylamine oxide



VI

To calculate the number of electrons in the outer shell (valence shell) of an atom in a molecule, all the shared (bonding) electrons (two for a single bond, four for a double bond and six for a triple bond) are counted for each of the bonded atoms because they surround both the bonded atoms. The unshared (nonbonding) electrons, if there are any, in an atom are counted only for that particular atom. Thus, in the structure VI written for trimethylamine oxide, the number of electrons calculated for the outer shell of nitrogen comes out to be ten (all shared) which is more than the maximum permissible number of electrons for the outer shell of the second-row elements (eight). This is why it is not considered as a valid Lewis structure of trimethylamine oxide.

The valid Lewis structure of trimethylamine oxide (shown above) carries a positive charge on nitrogen and a negative charge on oxygen. These charges are called **formal charges**. A formal charge is assigned to an atom in a Lewis structure by counting the electrons 'belonging' to that atom and comparing them with the valence electrons in the neutral atom. For counting the electrons belonging to an atom for the purpose of assigning a formal charge, the shared electrons are divided equally between the bonded atoms. Only half of the shared electrons are considered to belong to each of the bonded atoms. Unshared (nonbonding) electrons belong entirely to the atom in which they are found.

If the number of electrons belonging to an atom is more than its valence electrons, the atom is assigned a negative charge and if less than its valence electrons, then it is assigned a positive charge. If the number of the belonging electrons is equal to the valence electrons, then the atom is neutral and bears no charge.

Let us now determine the formal charge assigned to the atoms in the structure of trimethylamine oxide.

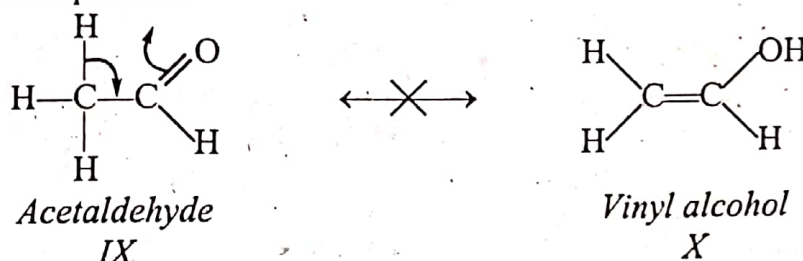
Atom	Valence electrons	Electrons belonging to the atom	Formal charge
H (each)	1	1	Zero
C (each)	4	4	Zero
N	5	4	+1
O	6	7	-1

So, in the structure of trimethylamine oxide, nitrogen is assigned a positive charge and oxygen a negative charge.

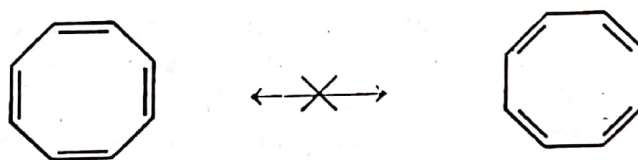
2. *The positions of the atomic nuclei must be the same in all the canonical forms.* The canonical forms of the same molecule differ only in the location of the electrons which are actually delocalized, e.g., VII and VIII.



The structures IX and X are not the canonical forms of the same molecule. In fact, these are two different compounds.



3. All atoms involved in the resonance must lie in the same plane. This is necessary to allow maximum overlap of the p orbitals for the delocalization of electrons. Cyclooctatetraene which has alternate single and double bonds, like benzene, does not show resonance similar to that shown by benzene (I and II).

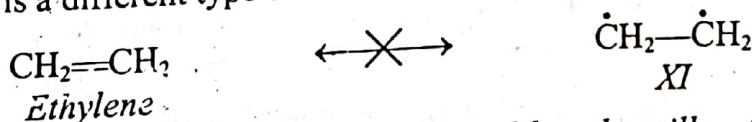


This is because its molecule is not planar but tub-shaped, as shown below, and the two p orbitals forming a double bond are not parallel to the p orbitals of the next double bond. This hampers delocalization of the π electrons.

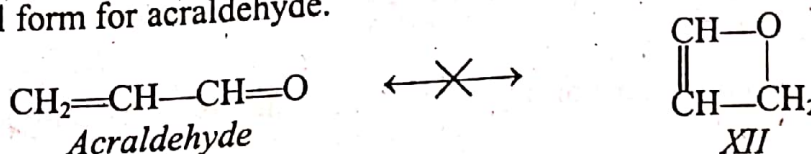


Cyclooctatetraene

4. The number of unpaired electrons must be the same in all the canonical forms. Thus, structure XI is not a valid canonical form for ethylene. It is rather an excited state of ethylene which is a different type of molecule.

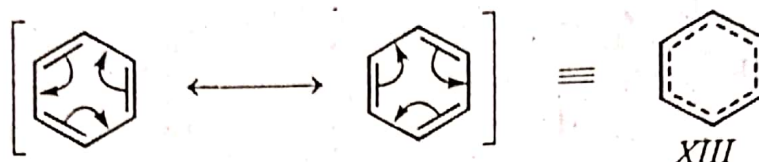


5. Structures with distorted bond angles or bond lengths will not contribute or will make little contribution to the resonance hybrid. Thus, the structure XII is not considered as a canonical form for acraldehyde.



Similarly, the structures III, IV and V make only a little contribution to the resonance hybrid of benzene because they have one bond each which is expected to be longer than a normal covalent bond.

In resonance, σ bonds are not involved and their position remains unchanged in all the canonical forms. It is only the π or unshared electrons which are placed at different positions in different canonical forms. Thus, if we write one canonical form for a molecule, then the other can be written by simply moving the π or/and the unshared electrons, as shown in structures VII and VIII or as shown below in the case of benzene (only the two main contributing structures are shown); the formal movement of electrons in resonance description, again, has no physical reality.



However, instead of writing the canonical forms, a representation like XIII can be used to express the resonance hybrid. In such representations, the bonds that are present in all the canonical forms, are drawn as solid lines while the bonds that are not present in all the canonical forms are drawn as dotted lines.

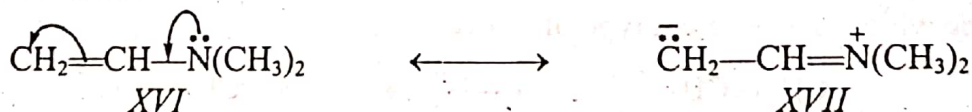
Contribution of canonical forms. All canonical forms do not contribute equally to the resonance hybrid. The contribution of each form corresponds to its stability. Since the resonance hybrid is more stable than any of the canonical forms, the more the stability of a canonical form, the closer it is in energy to the resonance hybrid and the more its contribution. It is not always easy to evaluate the relative stabilities of the canonical forms. However, certain rules may be helpful in this regard. These rules are given below and are self-explanatory.

Other things being equal,

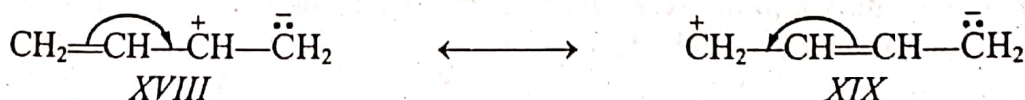
1. *The greater the number of covalent bonds, the greater the stability.* XIV is more stable than XV.



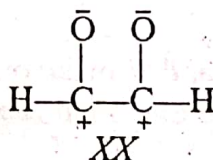
2. *Structures without a charge are more stable than those with formal charges.* XVI is more stable than XVII.



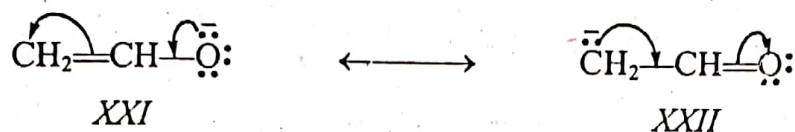
3. *The less the charge separation, the greater the stability.* XVIII is more stable than XIX. (A structure without formal charges means that there is no charge separation).



A structure with two like charges on adjacent atoms is highly unfavourable, e.g., XX.



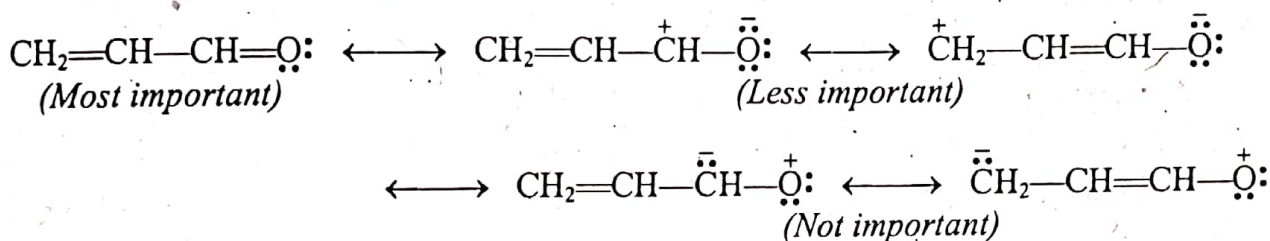
4. Structures with a negative charge on a more electronegative atom are more stable than those in which the negative charge is on a less electronegative atom. XXI is more stable than XXII.



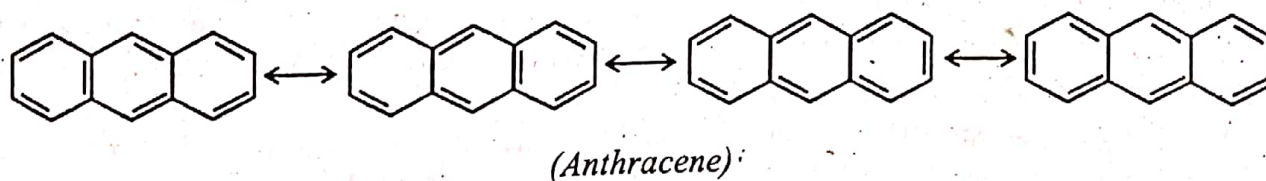
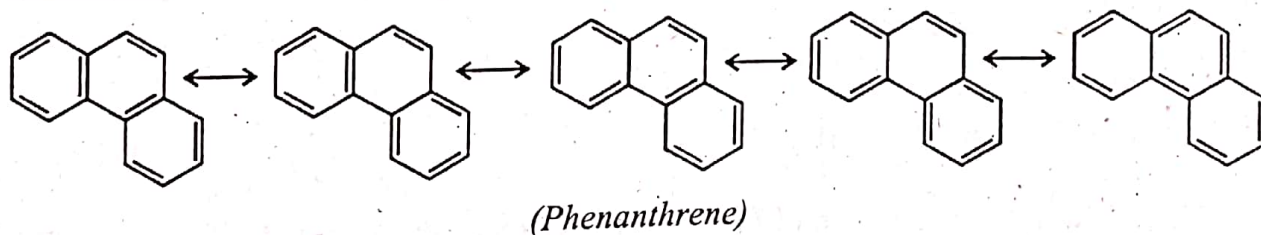
Similarly, positive charges are best carried on atoms of low electronegativity.

5. Equivalent canonical forms make equal contribution to the resonance hybrid. For example, I and II or $\text{CH}_2=\text{CH}-\overset{+}{\text{CH}}_2$ and $\overset{+}{\text{CH}}_2-\text{CH}=\text{CH}_2$.

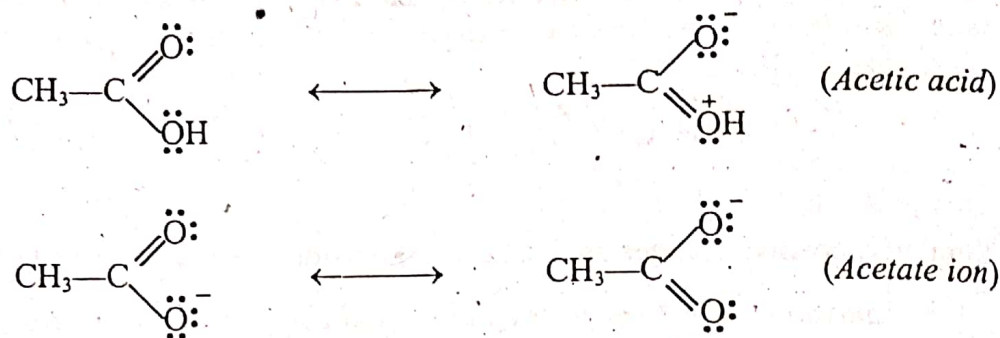
All the above rules are explained in the following example of acrolein in which the canonical forms are arranged in the decreasing order of their contribution to the resonance hybrid.



Stability of resonance hybrids. The energy of the resonance hybrid (actual molecule) is lower than that of any of the canonical forms. In other words, the resonance hybrid is more stable than any of the canonical forms. Resonance, therefore, is a stabilizing phenomenon and the hybrid is said to be resonance stabilized. The extent of resonance stabilization depends on the canonical forms that can be written for a molecule. In general, the greater the number of principal contributing structures of a molecule and more nearly equal their contribution, the greater the stability. For example, anthracene and phenanthrene both have three fused rings in their molecules but we can write five contributing structures, all equivalent, for phenanthrene, whereas only four such structures can be written for anthracene. Therefore, phenanthrene is more stable than anthracene.



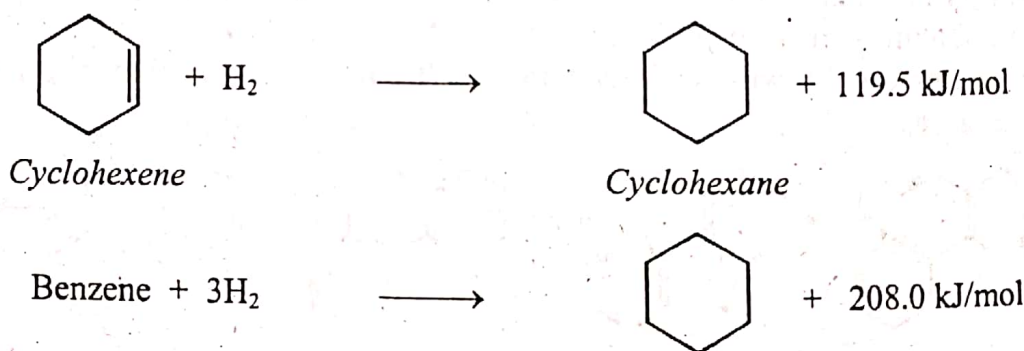
Similarly, we can write two canonical forms each for acetic acid and acetate ion.



In the case of acetic acid, one canonical form carries formal charges and therefore makes less contribution to the resonance hybrid than the other canonical form which is without a charge. On the other hand, both the canonical forms of acetate ion are equivalent and thus make equal contribution. Acetate ion is therefore more stable than acetic acid. This is why acetic acid readily loses proton (to be converted to a more stable anion) and thus behaves as an acid.

A quantitative measure of the resonance stabilization of a molecule may be obtained from the *resonance energy*.

Resonance energy. The resonance energy of a molecule is the difference between the energy of the most stable contributing structure and the energy of the actual molecule. Whereas we can determine the energy of the actual molecule practically, the energy of the most stable canonical form can only be estimated from its bond energies by comparison with an arbitrarily chosen model compound, because it does not actually exist. Numerical evaluation of the resonance energy of any molecule depends upon the system under consideration, the model chosen, the technique used and the assumptions made. For example, the resonance energy of benzene can be evaluated by measuring its heat of hydrogenation and comparing it with that of cyclohexene. In both the cases, the same product, i.e., cyclohexane, is obtained.



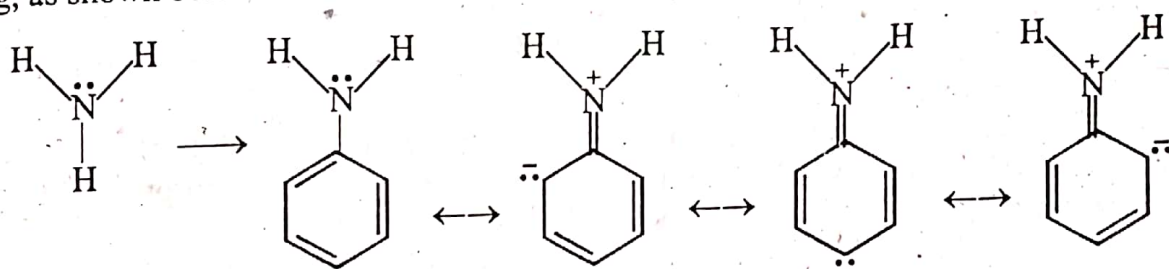
The heat of hydrogenation of cyclohexene which contains one $-\text{CH}=\text{CH}-$ group, is 119.5 kJ/mol. The most stable canonical form of benzene, i.e., cyclohexatriene (I) contains three such groups. It is, therefore, expected to release three times the heat of hydrogenation of cyclohexene ($119.5 \times 3 = 358.5$ kJ/mol). In actual practice, the heat of hydrogenation of benzene is 208.0 kJ/mol. Thus, the measured heat of hydrogenation of benzene is 150.5 kJ/mol ($358.5 - 208.0$) less than the expected value of the most stable canonical form. Benzene is, therefore, said to be resonance stabilized by an amount of 150.5 kJ/mol which is known as its *resonance energy*.

The resonance energy increases as the number of principal canonical forms increases. For example, the resonance energy of benzene, naphthalene, anthracene and

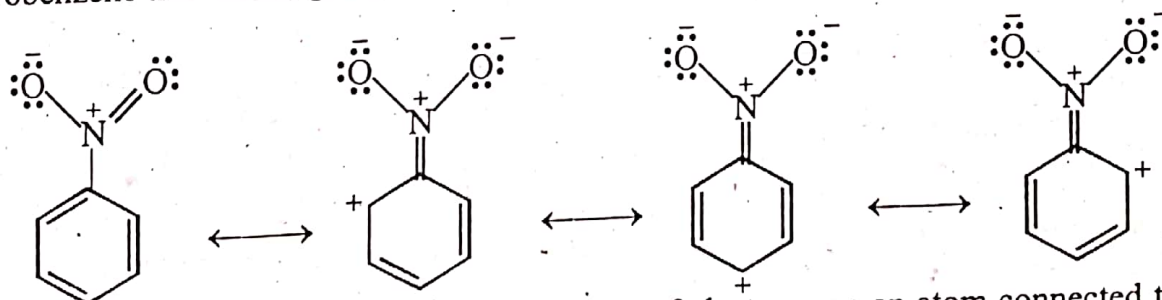
phenanthrene, for which we can draw two, three, four and five principal canonical forms, respectively, is 150.5, 255.0, 351.0 and 384.5 kJ/mol.

Resonance Effect

In a system involving resonance the distribution of electron density is different from the system that does not involve resonance. For example, in ammonia where resonance is absent, the unshared pair of electrons is located on the nitrogen atom. However, if one of the hydrogen atoms of the ammonia molecule is replaced by a benzene ring, the electron pair of nitrogen is delocalized over to the ring, resulting in the decrease of electron density on the nitrogen atom and the corresponding increase of electron density on the ring, as shown below:



This decrease in electron density at one position and the corresponding increase elsewhere is called the **resonance** (or **mesomeric**) effect. Thus, we say that the —NH₂ group in aniline donates electrons to the ring by the resonance effect. Similarly, in nitrobenzene the —NO₂ group withdraws electrons from the ring by a similar effect.



In general, a group containing an unshared pair of electrons on an atom connected to an unsaturated system, donates the electron pair to the unsaturated system, and a group containing a multiple bond, involving an electronegative atom, connected to an unsaturated system, withdraws electrons from the unsaturated system. The former group is said to be electron-donating (+M) and the latter group is said to be electron-withdrawing (−M) by the mesomeric effect. It must be emphasized that, as in the case of inductive effect (*p.* 20), the electrons are not actually being donated or withdrawn, rather the distribution of electron density is not the same as it would be in the absence of the resonance effect.

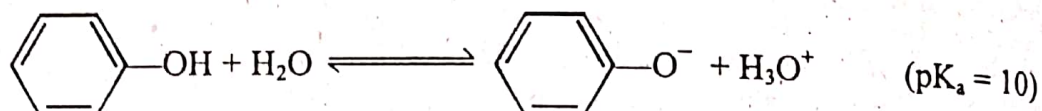
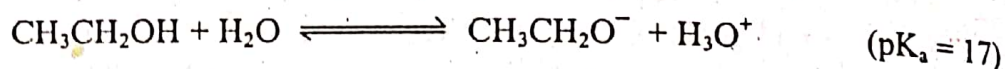
Some of the common +M and −M groups are listed below (not necessarily in the order of effect):

Electron-donating groups (+M): —O[−], —NR₂, —NHR, —NH₂, —OR, —OH, —OCOR, —I, —Br, —Cl, —F, —R, —Ar.

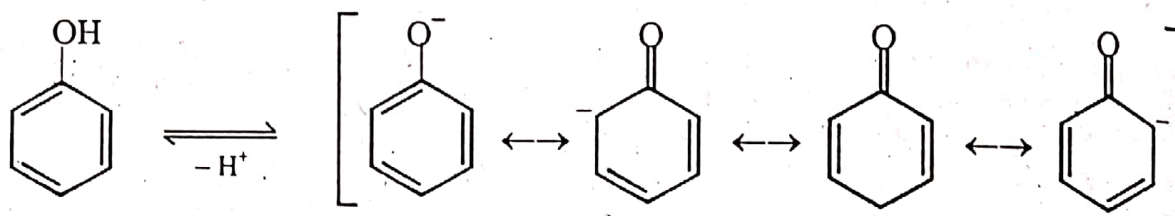
Electron-withdrawing groups (−M): —NO₂, —CN, —COOH, —COOR, —CONH₂, —CONHR, —CONR₂, —CHO, —COR, —Ar.

The group —Ar is capable of both kinds of effect. A group, whether +M or —M, causes resonance effect only when it is in conjugation with an unsaturated system because the resonance effect operates through π electrons.

The resonance effect has an appreciable influence on the physical characteristics and the chemical reactivity of organic compounds. Let us, for example, compare the acidity of phenol with that of ethanol. The acidity of both is a result of dissociation of the O—H bond, yet phenol is more acidic than ethanol by a factor of about 10^7 .

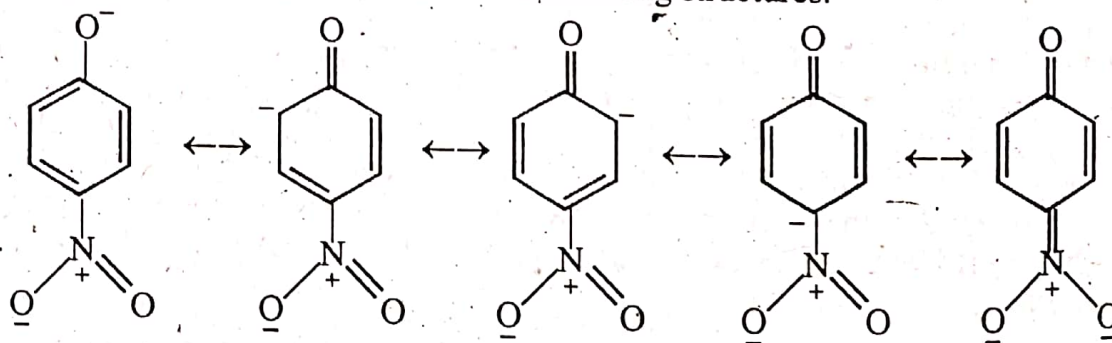


The enhanced acidity of phenol can be attributed to resonance stabilization through delocalization of the negative charge (electron pair) of the phenolate ion produced after the ionization of phenol.



This type of delocalization of the negative charge is not possible in ethoxide ion. Thus, phenolate ion is much more stabilized relative to phenol than is ethoxide ion relative to ethanol. Phenol therefore has much greater tendency to lose proton and behave as an acid as compared to ethanol.

Nitro group further enhances the acidity of phenol, particularly in the ortho and para positions, because it further delocalizes the negative charge over to the nitro group, increasing the number of the resonance contributing structures.



o-Nitrophenol and *p*-nitrophenol are approximately 1000 times more acidic than phenol itself, whereas *m*-nitrophenol in which the nitro group does not contribute toward resonance stabilization of the phenolate ion is only about 100 times more acidic than phenol (Table 2.1). The additional acidity of *m*-nitrophenol is due to the electron-withdrawing inductive effect of the nitro group. In fact, the enhanced acidity of *o*- and *p*-nitrophenols is due to the combined inductive and resonance effects of the nitro group. 2,4-Dinitrophenol is about 10^6 times more acidic than phenol.

Table 2.1
Acidity of Phenols

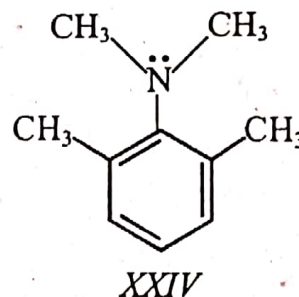
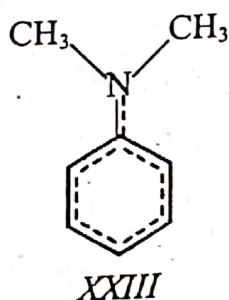
Phenol	pK _a
Phenol	10.00
<i>o</i> -Nitrophenol	7.23
<i>p</i> -Nitrophenol	7.15
<i>m</i> -Nitrophenol	8.35
2,4-Dinitrophenol	4.07

Similarly, the basicity of amines is very sensitive to resonance effect. For example, aniline is a weaker base than aliphatic amines by a factor of about 10^6 because the electron pair on the nitrogen atom which is responsible for the basic strength of the amines, is delocalized over to the aromatic ring in aniline (as shown at *p.* 37) and is not available for protonation to the same extent as is in the case of aliphatic amines where such delocalization is not possible. Nitro group further reduces the basicity of aromatic amines due to the reasons similar to those explained in the case of phenols.

Mostly, the resonance effect is observed alongwith the field effect (*p.* 22) and it is often difficult to separate the two effects. These are, therefore, frequently grouped together under the name of *electrical effects*.

Steric Inhibition of Resonance

As has been seen earlier, it is necessary for all the atoms involved in resonance to be in the same plane to facilitate delocalization of electrons. If some atoms are forced out of planarity, resonance is prevented or reduced. Consider, for example, the effect of ortho substituents on the basicity of N,N-dimethylaniline (XXIII) which is less basic than aliphatic amines due to delocalization of the unshared pair of electrons of nitrogen over to the aromatic ring, dimethylamino- group being coplanar with the aromatic ring. However, substitution of methyl groups at the ortho positions pushes the dimethylamino- group out of the plane of the ring, to ease off overcrowding, i.e., it inhibits coplanarity and thus prevents resonance. The unshared electrons of nitrogen are no more delocalized to the ring. This phenomenon is called **steric inhibition of resonance** and results in the increased basicity of 2,6,N,N-tetramethylaniline (XXIV) as compared to N,N-dimethylaniline. The basicity of XXIV becomes comparable to aliphatic amines.



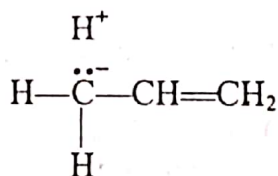
Hyperconjugation

So far we have discussed the delocalization of π or n electrons. There are also cases which involve delocalization of σ electrons. Consider, for example, the molecule of propylene in which C—H bond is in conjugation with an unsaturated system. The σ electrons of the C—H bond are delocalized and a canonical form such as XXV can be drawn.



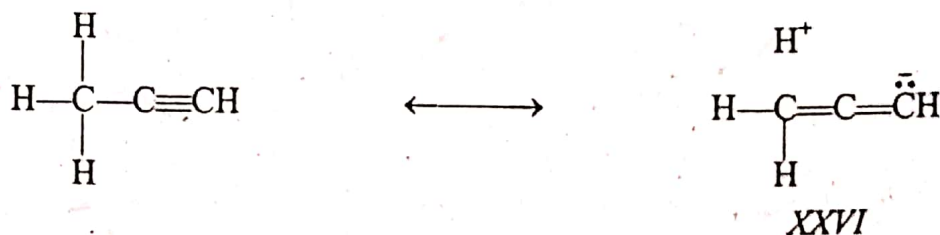
In this canonical form there is no bond between the carbon and the hydrogen atoms. This type of resonance is called **no-bond resonance**, and this system of conjugation where the σ electrons of a C—H bond are in conjugation with π electrons of an unsaturated system or with a p orbital, is known as **hyperconjugation**.

The hydrogen atom is not free to leave the molecule as it might appear from XXV which, in fact, does not exist but only contributes to the actual structure of the molecule. The effect of XXV on the actual molecule is that the electrons of the C—H bond are closer to the carbon atom than they would be if it did not contribute, and thus increase the ionic character of the C—H bond, as shown below:

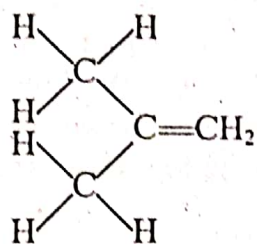


These electrons are partially delocalized. The delocalization in hyperconjugation ($\sigma-\pi/p$) is much less effective (about 10%) than in conjugation ($\pi-\pi/p$).

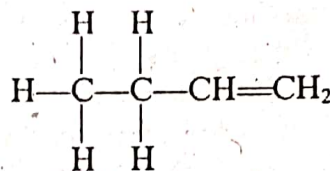
The concept of hyperconjugation was developed to explain certain physical data. For example, the observed CH_3-C bond length in methylacetylene is 1.46 Å which is shorter than the normal C—C single bond length (1.54 Å). This has been explained by a partial double bond character developed due to hyperconjugation, as shown in the canonical form XXVI.



Similarly, greater stability of isobutylene (XXVII) as compared to 1-butene (XXVIII), both having the same number of C—C, C=C and C—H bonds, determined on the basis of heats of combustion, has been explained by hyperconjugation.



XXVII

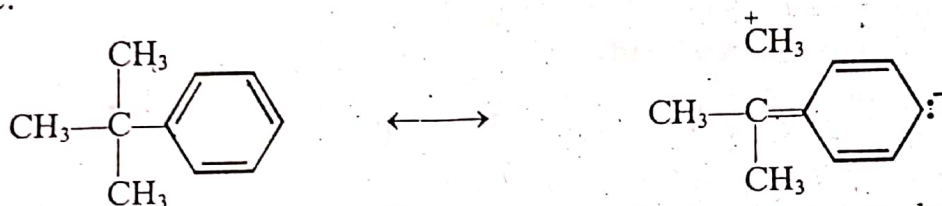


XXVIII

The former has six C—H bonds in conjugation with the unsaturated system as against two in the latter, and thus has greater number of canonical forms. This also shows that the effect of hyperconjugation becomes more pronounced as the number of α -hydrogen atoms increases.

The observed bond lengths and bond energies of neutral molecules, in the ground state, can also be explained by phenomena other than hyperconjugation, such as the state of hybridization of carbons, and hyperconjugation may not be operative in these molecules. However, hyperconjugation has certainly helped in explaining the stability of carbocations, free-radicals and the molecules in the excited states and in the transition states.

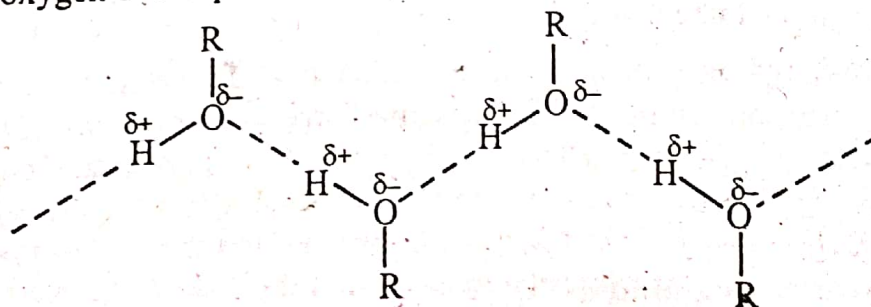
Evidences are also available for hyperconjugation involving C—C bond, of the following type:



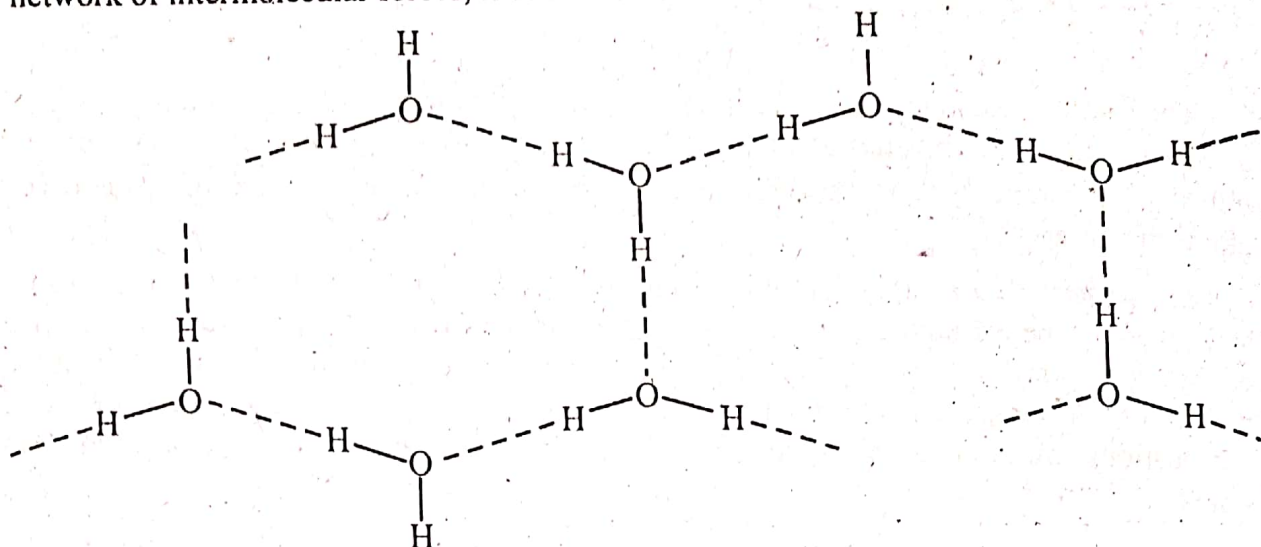
However, the hyperconjugation of C—C bond is generally less important than that of C—H bond.

Hydrogen Bonding

It has generally been observed that a compound containing hydrogen atom directly bonded to a highly electronegative atom like fluorine, oxygen or nitrogen, shows some unusual properties. For example, ethyl alcohol boils at 78°C , whereas boiling point of dimethyl ether is only -24°C , although both have the same molecular weight. The high boiling point of ethanol has been attributed to an *association* of its molecules which is due to some sort of dipole-dipole interaction (attraction between the positive end of one polar molecule and the negative end of another polar molecule) leading to the formation of a weak bond between the partially positively charged hydrogen atom of one molecule and the partially negatively charged oxygen atom of the other. (The partial charges on hydrogen and oxygen develop due to the large difference in their electronegativities).



Similar association is more pronounced in water which has still higher boiling point (100°C), although its molecular weight (18 g/mol) is much lower than that of ethanol (46 g/mol). (Methane with comparable molecular weight, i.e., 16 g/mol , is a gas). The aggregate of water molecules in which the molecules are linked to one another through a network of intermolecular forces, is shown below:



This type of weak bond (shown by dotted lines) between positive hydrogen and negative oxygen (or another electronegative atom) which is formed due to intermolecular forces is called **hydrogen bond**.

Since boiling of a liquid involves separation of molecules for transition to vapours, more energy, i.e., a higher temperature, is required to break the association. This is why the compounds forming hydrogen bonds, have higher boiling points.

The hydrogen bond, in a generalized form, can be expressed as $Z-H\cdots Y$ where Z is a small and strongly electronegative atom of low polarizability (ability of the electrons to be distorted; it depends on how loosely or tightly the electrons are held) like fluorine, oxygen or nitrogen, and Y is also one of these three atoms which are potential hydrogen bond acceptor by virtue of their unshared electron pair. In fact, the hydrogen atom bonded only to one of these three atoms is positive enough, and only these three atoms are negative enough to form a hydrogen bond. In hydrogen bonding, the hydrogen atom serves as a bridge between two electronegative atoms, holding one by a covalent bond and the other by a purely electrostatic force.

The geometry of $Z-H$ and Y is not changed by the hydrogen bond formation. For example, the $Z-H$ bond length remains unchanged during the formation of $Z-H\cdots Y$. The hydrogen is generally not equidistant from Z and Y . The $O-H$ bond length in alcohol is $0.96\text{ \AA}$, whereas the hydrogen-bonded $H\cdots O$ distance is $2.07\text{ \AA}$, i.e., about twice as large. However, the hydrogen bonding is more effective and thus more strong when the three involved nuclei, i.e., Z , H and Y are in a straight line.

Although the hydrogen bond is much weaker than an ordinary covalent bond, it is much stronger than the van der Waals attractive forces (forces between the molecules of nonpolar compounds). The strengths of the hydrogen bonds are in the range $8\text{--}42\text{ kJ/mol}$, whereas the van der Waals forces are in the vicinity of 4 kJ/mol ; the $O-H$ covalent bond dissociation energy in water is 464 kJ/mol . In general, the strength of the hydrogen bond increases with the acidity of hydrogen in $Z-H$ and the basicity of Y , as shown in Table 2.2.

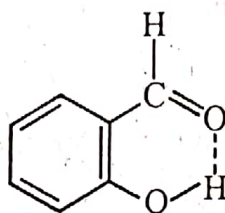
Table 2.2
Strength of Various Hydrogen Bonds

Nature of the hydrogen bond	Hydrogen bond energy
$\begin{array}{c} \text{H}-\text{F} \cdots \cdots \text{H}-\text{F} \\ \text{H}-\text{O} \cdots \cdots \text{H}-\text{O} \\ \qquad \qquad \\ \text{H} \qquad \qquad \text{H} \\ \qquad \qquad \\ \text{H} \qquad \qquad \text{H} \\ \text{H}-\text{N} \cdots \cdots \text{H}-\text{N} \\ \qquad \qquad \\ \text{H} \qquad \qquad \text{H} \end{array}$	41.84 kJ/mol 29.29 kJ/mol 8.37 kJ/mol

Fluorine, being more strongly basic and with a very low polarizability because of its electrons being close to and tightly held by the nucleus, forms a stronger hydrogen bond. In fact, the strongest hydrogen bond is $\text{F}^- \cdots \text{H}-\text{F}$ which has an energy of about 200 kJ/mol, and hydrogen is almost equidistant from the two fluorines.

If the hydrogen atom is too strongly acidic and the acceptor atom, Y , is too strongly basic, the hydrogen will shift as a proton to form a covalent bond with the acceptor atom, in a simple acid-base reaction.

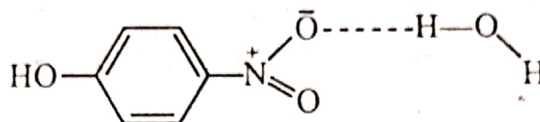
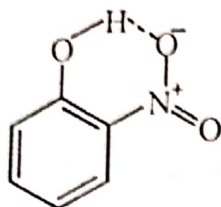
When $Z-H$ and Y are present in the same molecule, *intramolecular hydrogen bonding* is possible, which results in the formation of a ring. It is called **chelation** (holding of a hydrogen atom between two atoms of the same molecule) as against *association* that results from intermolecular hydrogen bonding. In most of the instances where chelation occurs, the hydrogen atom finds itself a member of a six-membered planar ring; the other five atoms of the ring share a π electron system, essentially equivalent to two double bonds. Salicylaldehyde presents a good example of chelation.



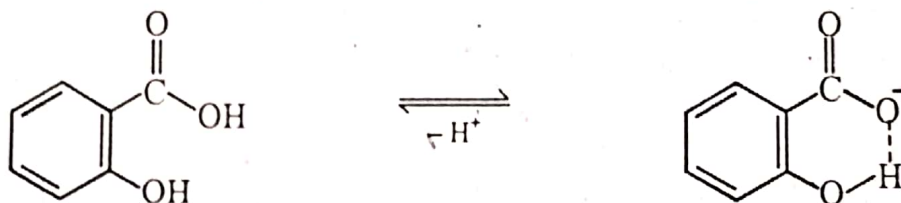
When a compound containing $O-H$ or $N-H$ cannot form an intramolecular hydrogen bonding, some degree of association through intermolecular hydrogen bonding which requires much less stringent steric requirements, almost always occurs.

Hydrogen bonding is important because of its effects on the properties of compounds. For example, association results in the higher boiling points than expected from the molecular weights of the compounds, as has already been explained. On the other hand, chelation usually causes lower boiling points than expected because in this case the molecule behaves as a monomer and is therefore easy to volatilize. Thus, the chelated salicylaldehyde boils at 196°C and can be steam-distilled readily, whereas its para isomer whose hydrogen bonding tendency must be satisfied only intermolecularly, boils above 240°C , and is not appreciably steam-volatile.

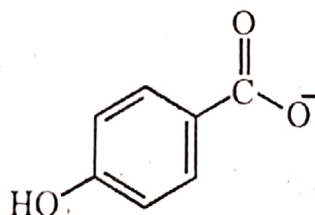
Similarly, hydrogen bonding affects the solubility of compounds. A vast majority of organic compounds that dissolve in water do so with an attendant formation of hydrogen bonds with the solvent. Thus, the solubility of *o*-nitrophenol in water is lower, as compared to its para isomer, because its hydrogen bonding with water molecule is restricted due to its intramolecular hydrogen bonding (which no doubt increases its volatility). On the other hand, *p*-nitrophenol is free to form hydrogen bonding with water, and thus is more soluble.



o-Hydroxybenzoic acid (salicylic acid) is approximately 40 times more acidic than *p*-hydroxybenzoic acid, because in the ortho isomer the hydroxyl group is in a better position to stabilize the carboxylate ion, formed after ionization, by chelation.



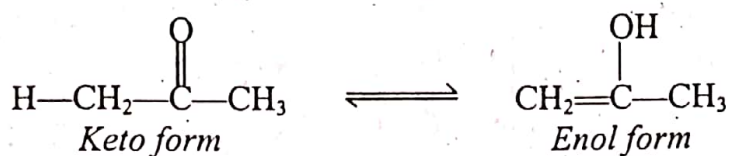
This facility is not available to the para isomer because the hydroxyl group is far away from the carboxylate ion.



Intramolecular hydrogen bonding is also responsible for the large amount of enol present in certain tautomeric equilibria (see p. 47).

Tautomerism

While writing the resonance structures to describe the actual structure of a compound it was observed that the relative positions of the atomic nuclei remained unchanged. The structures which differ in the relative positions of their atoms are not resonance forms but represent different compounds which are isomeric with each other and are capable of independent existence. If two isomers differ in the placement of a proton on one atom or another (mostly hetero atom such as oxygen or nitrogen) with the corresponding adjustment of a double bond, the phenomenon is referred to as **tautomerism** and the isomers are known as **tautomers**. The most common examples of tautomerism are provided by the carbonyl compounds containing α -hydrogen, in which the proton from the α -carbon atom shifts to the carbonyl oxygen, associated with the appropriate movement of electron pair, to form an enol (from *ene* + *ol*, i.e., unsaturated alcohol). This process is called **enolization**. The proton then moves back to convert the enol form back to the keto form.

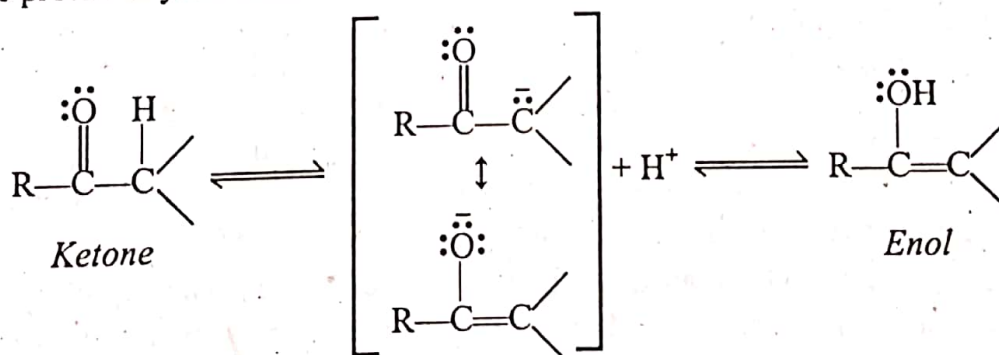


Thus, there is a rapid equilibrium between the keto and the enol forms. It is therefore also commonly known as **keto-enol tautomerism**. That both forms exist simultaneously is evident from the behaviour of acetoacetic ester.

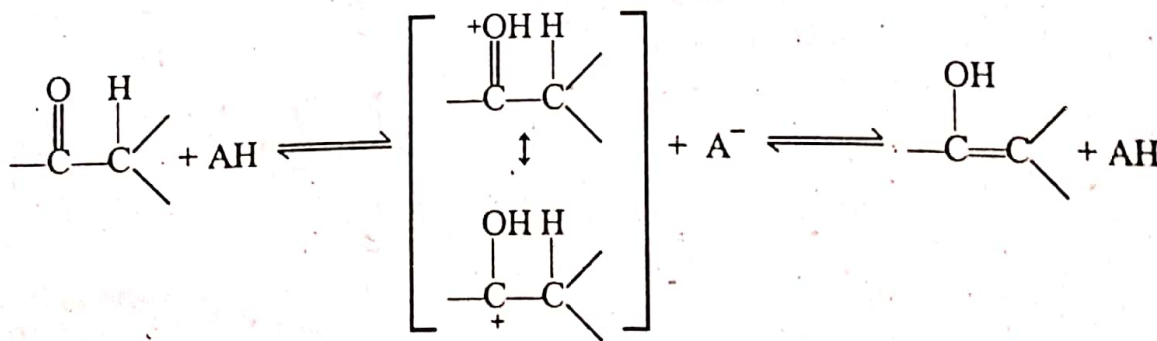
Acetoacetic ester forms oxime with hydroxylamine and cyanohydrin with hydrogen cyanide, indicating the presence of a keto group. On treatment with sodium it evolves hydrogen and forms sodium salt, indicating the presence of a hydroxyl group. It discharges the colour of ethanolic solution of bromine, indicating the presence of an olefinic double bond. It produces a reddish-violet colour on treatment with neutral ferric chloride, which is a characteristic test for enolic group ($-\text{C}=\text{C}-\text{OH}$). Thus, acetoacetic ester is, in fact, a mixture of keto and enol forms which are always in equilibrium with each other.

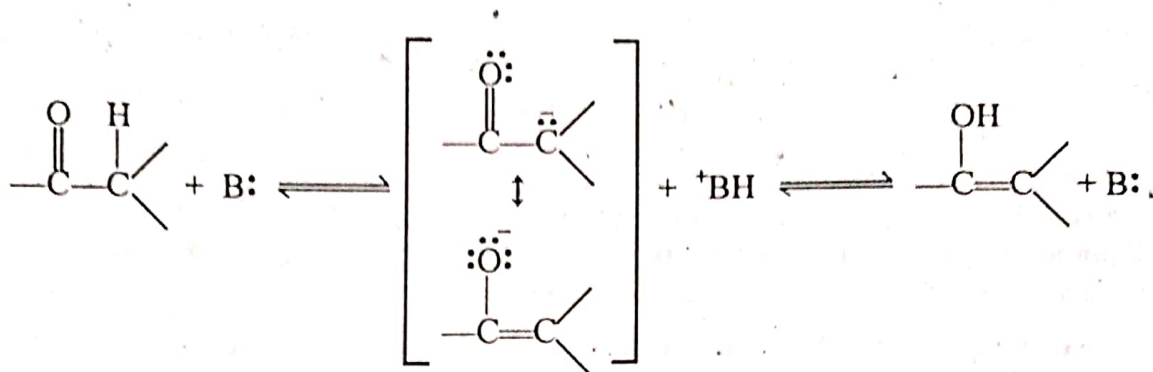


Either the ketone or its tautomeric enol may lose a proton and form a single intermediate, **enolate ion** (described by two resonance forms), which may recombine with the proton to yield either tautomer.

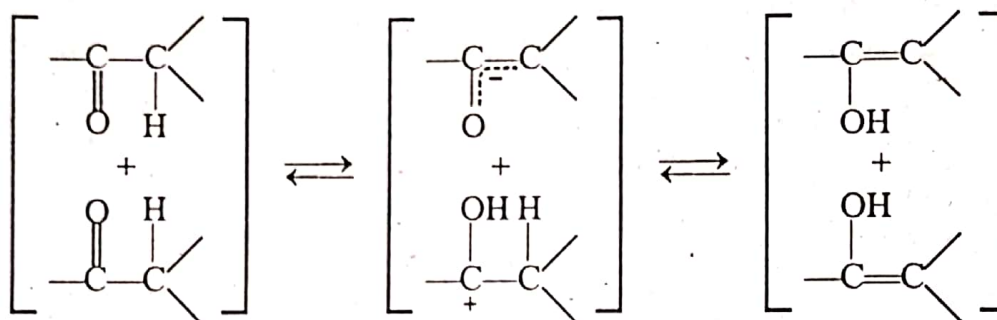


Since tautomerization (*interconversion of the tautomers*) involves the transfer of a proton, it is catalysed by both acids and bases.





In the absence of a catalyst, tautomerization occurs by autoprotolysis.

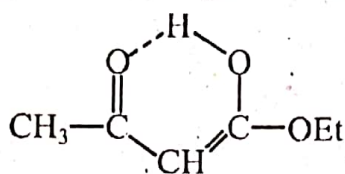


However, at -80°C and in the absence of a catalyst, either tautomer can be kept indefinitely.

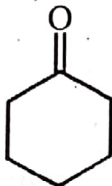
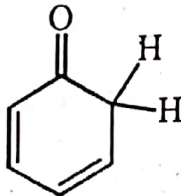
Since the two tautomers are readily and reversibly interconvertible by the appropriate movement of a proton, they cannot be separated except by special means. However, the position of equilibrium depends on the molecular structure. In simple cases of monocarbonyl compounds, such as acetone or acetaldehyde, the equilibrium lies well to the keto side because it is substantially more stable than its corresponding enol, and at equilibrium the enol content is very small. The relative stabilities (or energies) of the two tautomers may be roughly calculated from the bond energies of the bonds which undergo change on tautomerization. The keto form possesses a C—H, a C—C and a C=O bond as against a C=C, a C—O and an O—H bond for the enol form. The approximate sum of the bond energies of the former set of bonds for the keto form is 1502 kJ/mol and that of the latter set for the enol form is 1433 kJ/mol. The keto form is therefore more stable by about 69 kJ/mol. However, when the enolic double bond is a part of a conjugated system, the enol can be more stable than its corresponding ketone, due to resonance stabilization. For example, the possible keto tautomer of phenol is less stable by about 56 kJ/mol, as shown below, and is therefore almost unknown.

Bond energy	–1502 kJ/mol	–1433 kJ/mol
Resonance energy	–21 kJ/mol	–146 kJ/mol
Total energy	–1523 kJ/mol	–1579 kJ/mol
	Difference, $\Delta E = 56 \text{ kJ/mol}$	

Similarly, in certain molecules such as acetoacetic ester, the enol is also stabilized by intramolecular hydrogen bonding which is not possible in the keto form.



When the enol content is high, both forms can be isolated. The enol content of some of the carbonyl compounds at equilibrium is given below:

Carbonyl compound	Enol content, %
CH_3COCH_3	Traces*
CH_3CHO	0.1
$\text{CH}_2(\text{CO}_2\text{Et})_2$	0.8
$\text{CH}_3\text{COCH}_2\text{CO}_2\text{Et}$	8.0
$\text{CH}_3\text{COCH}(\text{CH}_3)\text{COCH}_3$	31.0
$\text{CH}_3\text{COCH}_2\text{COCH}_3$	76.0
$\text{C}_6\text{H}_5\text{COCH}_2\text{COCH}_3$	90.0
$\text{CH}_3\text{COCH}_2\text{CHO}$	100.0
	0.02
	100.0

* Less than 1 ppm.

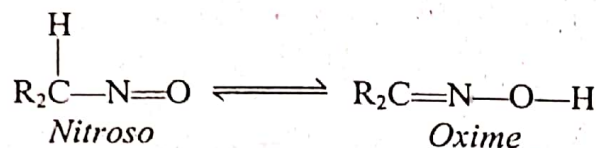
The extent of enolization is greatly affected by solvent. For example, the enol content of acetoacetic ester is 0.4% in water but 20.0% in toluene. Water stabilizes the carbonyl group by making hydrogen bond with it, rendering that group less available for internal hydrogen bonding through enolization. The effect of solvent on enolization is also clear from the enol content of acetylacetone in different solvents, as shown below:

Solvent	Enol content of acetylacetone, %
Water	15
Acetonitrile	58
Pure liquid (no solvent)	80
Hexane	92

Enolization is also affected by concentration and temperature.

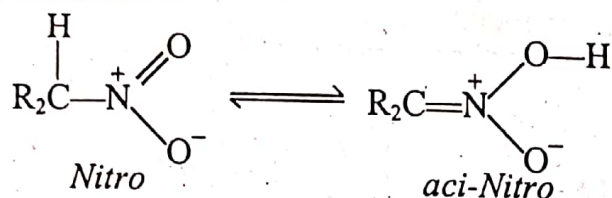
There are certain other systems that involve a rapid shift of proton to show tautomerism, similar to the keto-enol tautomerism. For example,

1. Nitroso-oxime tautomerism.



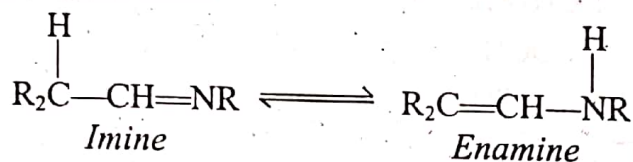
Oximes are more stable than the nitroso compounds.

2. Nitro-*aci*-nitro tautomerism.



Nitro form is more stable than *aci*-nitro form.

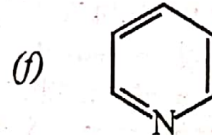
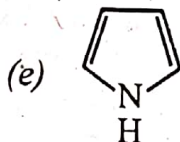
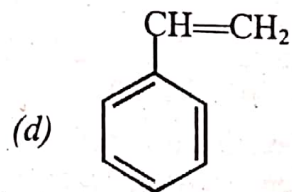
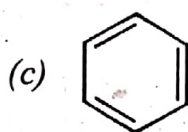
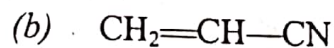
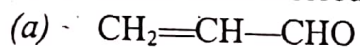
3. Imine-enamine tautomerism.



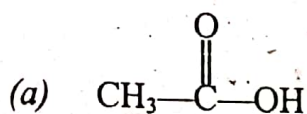
Imine is more stable than enamine.

Exercises

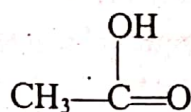
1. Draw all the possible molecular orbitals for each of the following molecules and arrange them in the order of their increasing energy. Also indicate the number of electrons in each molecular orbital, in the ground state.

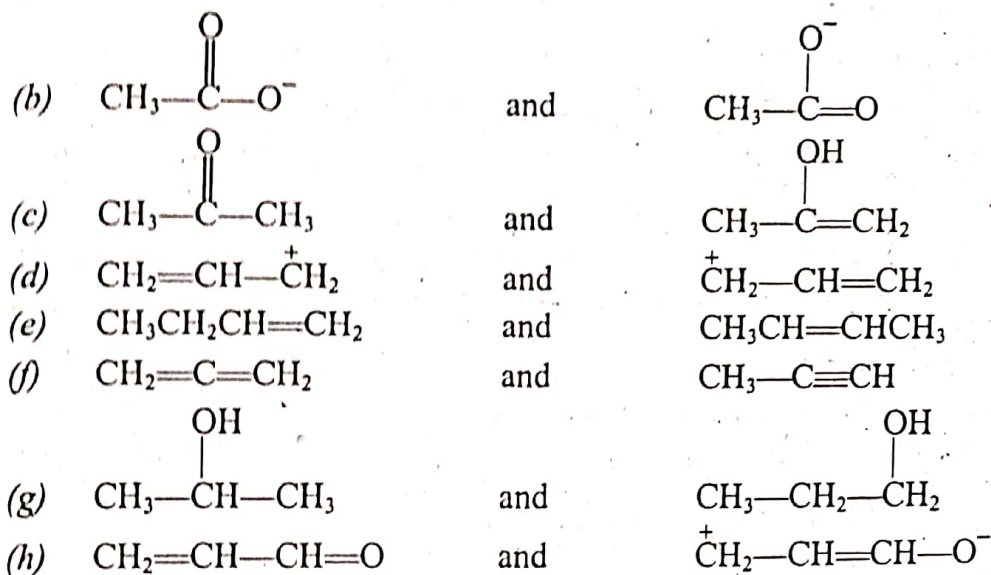


2. Which of the following pairs of structures are canonical forms of the same resonance hybrid? Describe the mutual relationship of the pairs which do not contribute to the same resonance hybrid.

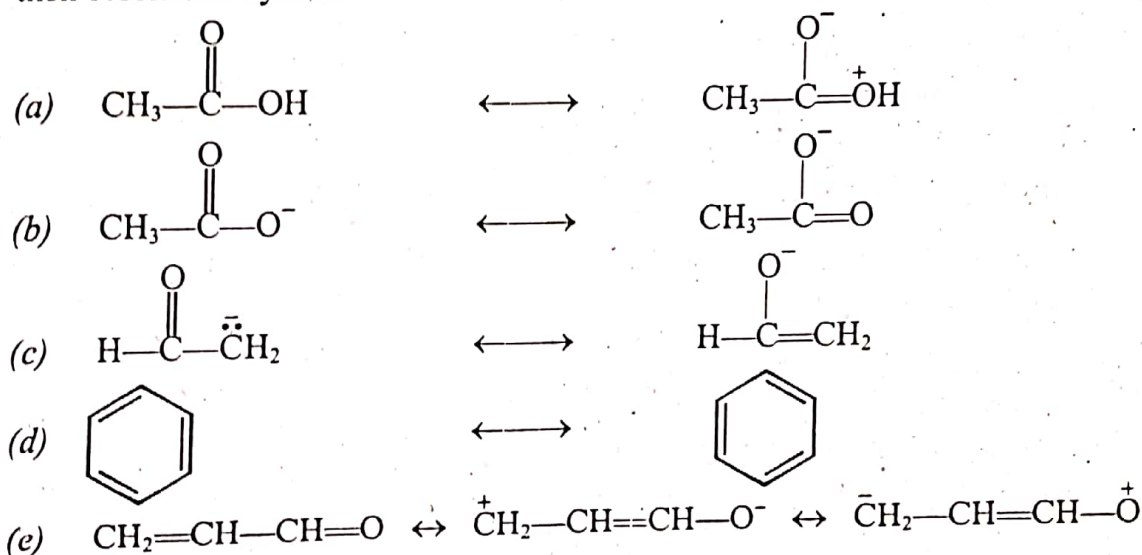


and

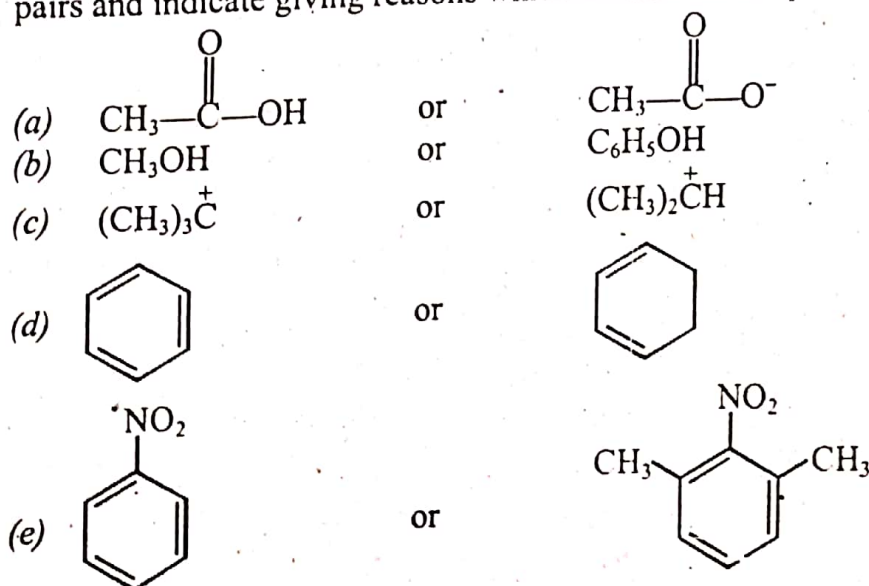


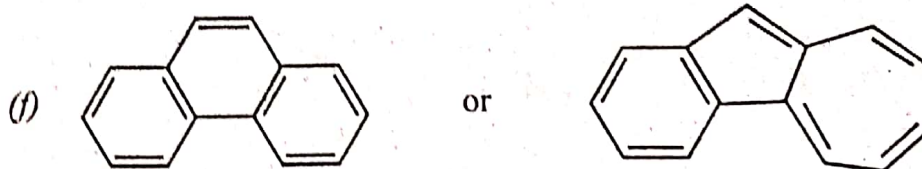


3. Compare the relative contribution of the canonical forms given below toward their resonance hybrid.



4. Draw the resonance contributing structures for each member of the following pairs and indicate giving reasons which member of the pair is more stable.





5. Calculate the resonance energy of cyclohexadiene. Heats of hydrogenation of cyclohexene and cyclohexadiene are -119.5 kJ/mol and -231.5 kJ/mol, respectively.
6. Dipole moment of 2,4,6-trimethylnitrobenzene (3.65 D) is closer to that of nitromethane (3.50 D) than to that of nitrobenzene (3.95 D). Explain.
7. Which of the following compounds would involve hydrogen bonding and what type of hydrogen bond (intermolecular or intramolecular) would they form?

(a) Acetic acid	(b) Acetaldehyde
(c) Ammonia	(d) Ethane
(e) Ethanol	(f) Ether
(g) Fluoroform	(h) Oxalic acid
(i) Salicylaldehyde	(j) Acetylacetone
(k) <i>p</i> -Hydroxybenzaldehyde	
8. The electronegativity of nitrogen and chlorine is almost the same, yet chlorine does not form hydrogen bond whereas nitrogen does. Explain.
9. Explain the difference between:
 - (i) Inductive effect and resonance effect.
 - (ii) Localized bond and delocalised bond.
 - (iii) Conjugation and hyperconjugation.
 - (iv) Resonance and tautomerism.
10. Explain why?
 - (i) Ethanol has higher boiling point than diethyl ether.
 - (ii) Methylamine is more basic than aniline.
 - (iii) Tertiary carbocation is more stable than primary carbocation.
 - (iv) Acetoacetic ester gives ferric chloride test which is a typical test for phenols.